

Algorithms useful for calculating multi-component equilibria, phase diagrams and other kinds of diagrams

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ABSTRACT

Phase diagrams are important in materials science. They consist of lines separating regions with different sets of stable phases and different kind of axes can be used. Many available phase diagrams are drawn directly from experimental information. However, they are closely related to the thermodynamic properties of the phases and can be calculated from thermodynamic models of the phases using the Calphad technique. This paper presents a set of algorithms to calculate equilibria and several kinds of diagrams using a very flexible set of conditions and axes. The algorithms can be applied to multi-component systems using model parameters in thermodynamic databases.

1. Introduction

Phase diagrams consist of lines separating regions with different sets of stable phases at thermodynamic equilibrium. Different axes can be used, most frequently temperature and composition but also chemical potentials or other properties. Phase diagrams are used quite differently in chemistry, physics and materials science. In this paper, most references will be to the latter.

The book published by Kaufman and Bernstein [1] in 1970, established the Calphad method to calculate phase diagrams by introducing a unified set of so called lattice stabilities for the pure elements. These describe the Gibbs energy of the pure elements in phases for which the element itself is not stable relative to the stable phase. Based on these, a large number of binary, ternary and higher order systems has been assessed using experimental and theoretical data on phase diagram and thermodynamic properties. The use of a unified set of lattice stabilities has made it possible to create consistent thermodynamic databases. During the last 50 years, several multi-component databases, with gradually more sophisticated models, have been developed together with software able to calculate phase diagrams as well as to simulate phase transformations when combined with kinetic data.

The databases are prerequisite for a software to calculate multi-component equilibria and phase diagrams. Today the possibility to calculate phase diagrams is a basic tool for a material engineer but the databases used are often encrypted and the thermodynamic models and the softwares are black boxes.

There are several books and papers explaining phase diagrams including some advice how to calculate them. Recently published books are by Pelton [2] and Hillert [3], both of which also explain thermodynamic modeling and its use for simulation of materials transformations. A book devoted to modeling and database assessments was published by Lukas et al. [4]. A general but a bit old introduction to the Calphad method was published by Saunders and Miodownik [5]. A summary of problems calculating phase diagram from databases was published by Chang et al. [6] and a recent paper about the importance and use of phase diagrams was published by Schmid-Fetzer [7].

An algorithm able to calculate equilibria using as condition that a phase is stable with zero amount together with a method to change the condition values along the axes of the diagram is known as mapping. The implementation of this in the free OpenCalphad (OC) software [8] available at [9] is explained in this paper. In the Thermo-Calc (TC) software [10] this technique had been developed by Bo Jansson [11] even before Gupta et al. [12] formulated the Zero Phase Fraction (ZPF) concept. Both in OC and TC it is possible to plot the calculated ZPF lines with other axes than those used for the calculation. Other commercial software which can calculate multi-component phase diagrams have not published their algorithms.

In the following Section 2, a few terms and concepts are defined or explained and the way to calculate a single equilibrium with a very flexible set of conditions is introduced. How to perform calculations is described in Section 3 and in Section 4 different ways of presenting the calculated results will be discussed.

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2. Computational thermodynamics basics

To calculate an equilibrium in a system with n components, we must set $n + 2$ thermodynamic variables as conditions. Frequently the temperature, T , the pressure, P , and the number of moles of each component A, N_A , are used but many others are possible, for example prescribing a stable phase.

The way to define the amounts and the compositions of the chemical species involved in a system can be a source of confusion. In this section, some terms and quantities used later will be defined. The basis of the thermodynamic models will then be introduced as well as the upmost important issue within our work that is the way how a single equilibrium is calculated. Finally we will discuss some basic features on calculated phase diagrams *i.e.* the objects that the algorithms that we will present in Section 3 have to be able to calculate.

2.1. Composition variables

Handling the composition of a system requires some prior definitions.

1. An **element** is an atom in the periodic chart.
2. A **species** is a molecular like aggregate of atoms with fixed stoichiometry, for example H_2O . For the modeling, its stoichiometry can be a real number as for example in $CuS_{0.5}$. The values 2 and 0.5 are called stoichiometric factors and denoted b_{iA} where i is the species containing the element A. The elements are the simplest species. A species can have a net charge, *i.e.* be an ion. The vacancy, denoted Va, is a special species that does not take part to the mass balance but contributes to the configurational entropy by occupying sites not occupied by any other species.
3. A **constituent** is a species in a phase. It can be experimentally observed or a modeling tool. In a phase α , the fraction of the sublattice s occupied by the constituent i is denoted y_{is}^α . For each sublattice,

$$\sum_i y_{is}^\alpha = 1 \quad (1)$$

4. A **component**, sometimes called system component, is normally an element. It can be convenient to define another orthogonal set of species as components, for example Al_2O_3 -CaO-SiO₂-O instead of Al-Ca-O-Si. The number of components must be the same as the number of elements. In this paper, we will only consider cases where components are the same as the elements.

The number of moles of a component A per formula unit of a phase α is calculated as:

$$M_A^\alpha = \sum_s a_s^\alpha \sum_i b_{iA} y_{is}^\alpha \quad (2)$$

the mole fraction of A in α is:

$$x_A^\alpha = \frac{M_A^\alpha}{\sum_A M_A^\alpha} = \frac{M_A^\alpha}{M^\alpha} \quad (3)$$

The number of moles of A per formula unit of the phase, M_A^α , is used later in Eq. (6) to express the mass balance.

2.2. Thermodynamic models

The property modeled in Calphad is the Gibbs energy for each phase separately. The generic expression for the Gibbs energy of a phase α is:

$$G_M^\alpha(T, P, y_{is}^\alpha) = {}^{srf}G_M^\alpha - T {}^{cfg}G_M^\alpha + {}^EG_M^\alpha + {}^{phy}G_M^\alpha \quad (4)$$

where ${}^{srf}G_M^\alpha$ is the surface of reference for the phase, T the absolute temperature, ${}^{cfg}G_M^\alpha$ the configurational entropy, ${}^EG_M^\alpha$ the excess Gibbs energy and ${}^{phy}G_M^\alpha$ the contribution from special physical phenomena

such as ferromagnetic transitions for the phase α . The set of constituent fractions, y_{is}^α , can be different for each phase depending on the model used.

Note that the subscript M on G_M indicates that the properties are per mole formula unit of the phase because the amount of the components may vary if the phase has vacancies or molecules as constituents, the notation G_m is reserved for the Gibbs energy per mole of components.

The Gibbs energy for a system with one or several components A and one or more stable phases α is:

$$G(T, P, N_A) = \sum_\alpha \aleph^\alpha G_M^\alpha(T, P, y_{is}^\alpha) \quad (5)$$

where $\aleph^\alpha \geq 0$ is the amount of formula units of α . At equilibrium the Gibbs energy is minimum for a system with given values of T, P and N_A .

2.3. Single equilibrium calculations

2.3.1. The Lagrange multiplier method

Most algorithms for equilibrium calculations use the Lagrangian multiplier method to eliminate constraints and conditions. Partial derivatives of this is used to find the minimum of the Gibbs energy. In OC, it is implemented following a proposal by Hillert [13] and first used by Jansson [11] in the POLY module of Thermo-Calc software [10].

$$L = \sum_\alpha \aleph^\alpha G_M^\alpha(T, P, y_{is}^\alpha) + \sum_A \mu_A \left(\sum_\alpha \aleph^\alpha M_A^\alpha - \tilde{N}_A \right) + \sum_\alpha \sum_s \eta_s^\alpha \left(1 - \sum_i y_{is}^\alpha \right) + \sum_\varphi \gamma^\varphi \aleph^\varphi \quad (6)$$

where the summation over α are for all phases and the summation over φ is for all metastable phases. μ_A , η_s^α and γ^φ are Lagrangian multipliers. The function L will have the same extremum as $G(T, P, N_A)$ when all constraints are fulfilled. The constraints here correspond to the total number of moles for each component, $\tilde{N}_A$, the fact that the sum of fraction on each sublattice equals unity, Eq. (1), and the fact that the amount of the metastable phases, $\aleph^\varphi$, equals zero. More constraints for phases with ions or if P is not constant can be added. Eq. (6) can be solved by a linear iterative method explained by Hillert [13].

An important feature of Eq. (6) is that the Lagrangian multiplier μ_A is the chemical potential because at the minimum we have:

$$\frac{\partial L}{\partial \aleph^\alpha} = 0 = G_M^\alpha + \sum_A \mu_A M_A^\alpha \quad (7)$$

for each stable phase α .

Another interesting feature in Eq. (6) is that the multiplier γ^φ is the driving force of φ . At equilibrium, it is negative for a metastable phase. The last term in Eq. (6) makes it possible to handle the change of the set of stable phases during the iterations. A positive driving force γ^φ means that the metastable phase φ is to be added to the set of stable phases. If a stable phase α has negative amount, $\aleph^\alpha$, at an iteration, it is removed from the set of stable phases.

Fig. 1 presents the flowchart of the algorithm to calculate a single equilibrium with different sets of conditions, referred in this paper as algorithm A. It is used by the algorithms presented later to calculate diagrams with varying conditions. It is a linear algorithm because it uses second derivatives of the Gibbs energy to convert the conditions on extensive or normalized variables to be functions of the chemical potentials and the phase amounts. The details of the calculation in OC is described in [14].

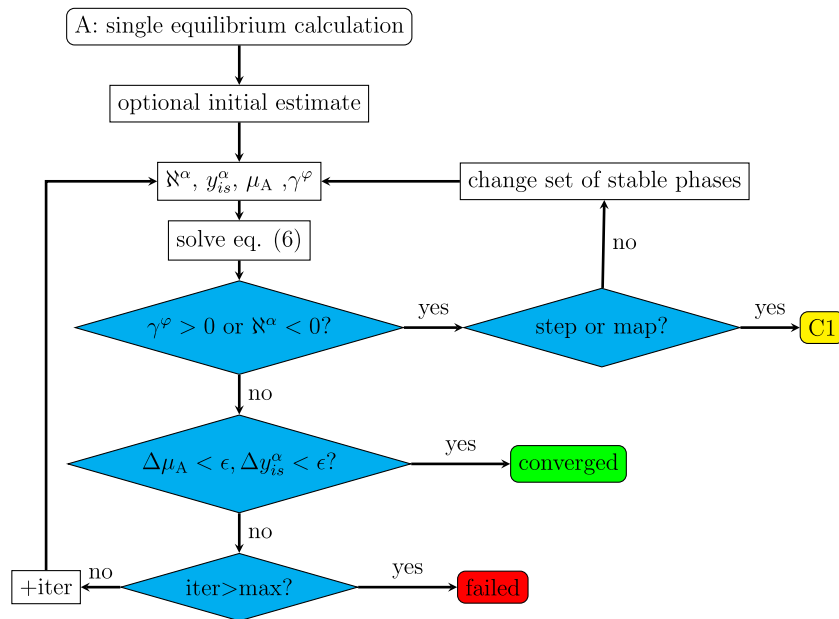


Fig. 1. Flowchart of algorithm A to calculate a single equilibrium.

2.3.2. Conditions for an equilibrium calculation

The Gibbs phase rule requires that $n + 2$ conditions are given to calculate an equilibrium, in a system with n components. In most cases, T, P and the overall amount or fraction of the components. Using such a set of conditions, which represents a closed system, the equilibrium will be found by a minimum in the Gibbs energy. But a system can be defined in many other ways by exchanging one or more of the $n + 2$ conditions as listed below. Using other conditions than T, P and N_A is taken care of by Lagrange constraints in the minimizer. We can use:

- Conditions on the chemical potential or activity of a component. With such a condition the system is “open” as the amount of that component is not specified. It is also possible to set conditions on both amount and chemical potential of a component if, for example, some other component has neither amount nor potential set.
- Conditions on entropy, S , or enthalpy, H , for heat balance calculations.
- Conditions on the volume, V provided that the database has volume description for the phases.
- Specifying that a phase must be stable with a given amount, for example zero.
- Conditions on the mole fraction of a component in a phase, x_A^α , to find a solubility limit.
- Conditions on constituent fractions, y_{is}^α , to calculate a second order transition.
- Conditions that are an expression of state variables, for example $x_{Cr}^{bcc} - x_{Cr}^\sigma = 0$ to calculate the congruent transformation of σ to bcc.

Each condition is set separately and most of these conditions can also be used as an axis variable for calculating equilibria along a line.

2.3.3. Global equilibrium calculation

In Fig. 1, algorithm A starts by an initial estimate of the stable phases and their constitution. This estimate can be the results of a previous calculation, provided by the user or by an automated procedure. In OC, a grid minimizer is used to find this automated set [14]. The grid minimizer approximates the Gibbs energy surface of each phase with a set of gridpoints and selects the set that gives the lowest Gibbs energy for the current conditions. The phases and constitutions that these

selected grid points represent are used as starting values by algorithm A to solve Eq. (6). The grid minimizer can find the global equilibrium and detect phases with miscibility gaps. A phase with a miscibility gap has a single Gibbs energy function but can appear as two (or more) separate phases with different compositions in the calculations and diagrams. In OC and TC they are identified with a suffix #<digit>. The fact that the phases are modeled with a single Gibbs energy function does not change Gibbs phase rule. The grid minimizer will also set the constitution of the metastable phases using a gridpoint where its Gibbs energy is closest to the Gibbs energy tangent plane of the selected gridpoints.

If the set of conditions does not allow the use of the grid minimizer, for example if T is not a condition, an initial set will be guessed and after the equilibrium calculation, when T and the overall composition has been calculated, the grid minimizer can be used as test and the equilibrium is recalculated automatically if there are any gridpoints below the calculated equilibrium.

During the line calculations, explained in Section 3, global checks are made at node points and at regular intervals along a line. If such a check finds that there is another set of phases which represent a more stable equilibrium, the automatic procedure is to abandon this line and suppress it in a subsequent plot. This can typically happen if a stable phase would like to separate into a miscibility gap when calculating along a line.

2.4. Calculated phase diagrams

Most phase diagrams in materials science are at constant pressure. Binary isobar “ $T - x$ ” diagrams can be plotted as shown in Fig. 2. In such diagrams, a line is either a solubility line or part of an invariant equilibrium with fixed values of the potentials but with varying amount of the phases. As the tie-lines are in the plane of the presentation, the composition of the phases and their amounts can easily be read from the diagram. Not only stable phase diagrams, as in Fig. 2.a and b, but also metastable ones, as in Fig. 2.c, can be calculated.

Ternary diagrams require that some additional system variable is fixed, as for example the temperature for an iso-thermal diagram shown in Fig. 3(a). In the case of iso-thermal sections of ternary systems also, tie-lines are in the plane of representation. In Fig. 3(a), they are drawn as light green lines connecting compositions where two phases are in equilibrium. The green triangles define 3-phase regions; their corners

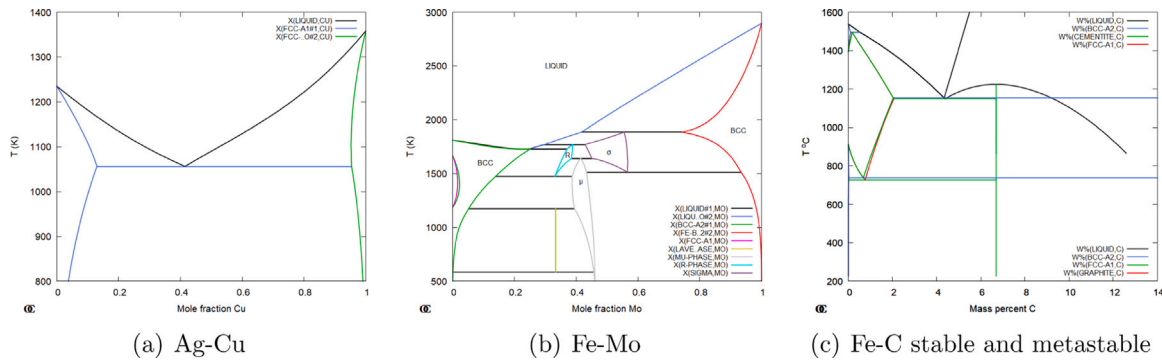


Fig. 2. (a) Ag-Cu phase diagram calculated using Hayes et al. [15] description. (b) Fe-Mo phase diagram calculated using SGTE SSOL database [16]. (c) Fe-C phase diagram calculated using the same database overlaying stable states with graphite and metastable with cementite.

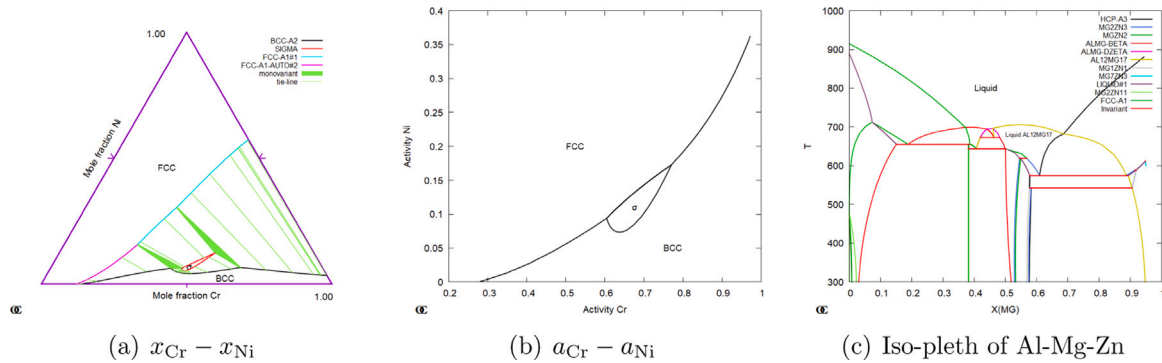


Fig. 3. (a) Cr-Fe-Ni iso-therm at 1000 K calculated using SGTE SSOL database [16] plotted as a Gibbs triangle. (b) Results from the same calculation plotted using the chemical potentials of the elements. (c) Iso-pleth in the Al-Mg-Zn system at $x_{Zn} = 0.05$ calculated using the COST 507 [17] database.

indicate the compositions of the three phases in equilibrium. From such a diagram, the composition and the amount of each phase can be read. In Fig. 3(b), the results from the same iso-thermal calculation as in (a) is plotted using the chemical potential of the components as axes. The two-phase regions become lines and the three-phase regions points because the elements in the stable phases have the same chemical potentials. In this case all the regions are single phase. The σ phase area seems like a balloon on the line separating fcc and bcc.

If one or more compositions are kept constant, an iso-pleth section is obtained as shown in Fig. 3(c) for the Al-Mg-Zn system at $x_{Zn} = 0.05$. An iso-pleth section can actually also be defined by a constant ratio between two elements or more generally by a linear equation between several compositions. In Fig. 3(c), the lines separate regions with different sets of stable phases. The horizontal lines are invariant equilibria with 4 stable phases. In such a section, there are no tie-lines. The phase amounts and compositions are not shown and thus cannot be read from such diagram except in the areas where a single phase is stable. By using a database such information can readily be obtained with additional calculations for the interesting conditions.

3. Calculating lines with varying conditions

A software which can calculate equilibria can be extended to calculate equilibria along a line by varying a single condition between some limits to find how the properties of the system changes with this condition, this is a **step** calculation. The procedure to perform such calculations follows the left branch in the algorithm B in Fig. 4 and will be discussed in more details in Section 3.2.

Using more axes, it is possible to calculate a phase diagram. These latter calculations will be made following Zero Phase Fraction (ZPF) lines, a concept discussed in [12,18]. ZPF lines separate regions in a phase diagram where a phase is present from regions where it is not present. A phase diagram with at least one composition variable

consists entirely of ZPF lines. The **mapping** procedure will replace one axis condition by a condition that a phase is stable (fix) with zero amount. It will be discussed in more detail in Section 3.3.

Both stepping and mapping will thus consist in calculating lines. The basic operations to perform these two kinds of diagram calculations in the OC software are illustrated in the general algorithm B in Fig. 4 where both branches of this general algorithm end in calling the algorithm C1 in Fig. 5 whose task is to calculate a line.

As shown in algorithm B, to calculate diagrams, the simplest way to start is to set the appropriate conditions for a single equilibrium calculation and then select one or more conditions as axis variable with a minimum and maximum value and a maximum increment between each calculation.

In some cases, a single start point may not find all the lines in the phase diagram. The Fe-Mo phase diagram shown in Fig. 2(b) is one of these. The γ loop close to pure Fe is not connected to the other lines of the diagram. Depending on the starting conditions, either the loop or the rest of the diagram will not be found by the basic algorithms presented hereunder. More than one starting point has then to be input. This can simply be done by the user or it is possible to automatically predefine many starting points, in particular for binary or ternary diagrams. Such issues will not be considered in the algorithms presented here, where instead the focus is on the way to process connected lines.

3.1. Data structures

The design of the data structure to solve a computational problem is of great importance for software development. With a poor data structure the coding becomes more complex and difficult to maintain.

The calculations are organized by following **lines** of single equilibria with the same set of stable phases by incrementing one of the axis conditions in algorithm C1 in Fig. 5.

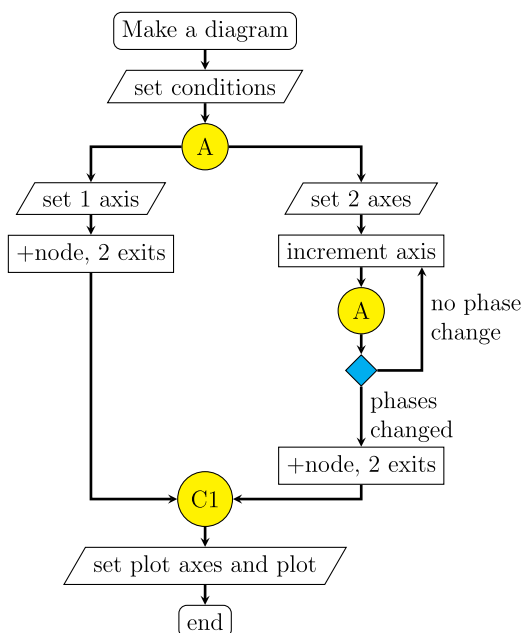


Fig. 4. Algorithm B: To calculate and plot a diagram in OC.

A **node** point is where a line starts or ends. They are created from the start equilibria provided by the user or by algorithm C2 when the set of stable phases changes along a line and the line is terminated. All nodes contain **exits** which will be used as a start of a line to be calculated. An exit will be marked as done if it is identical to the end of a calculated line terminating at a node which already exists. Each exit has a complete description of the equilibrium at the node, *i.e.* T , P , the amount and constitution of all phases and the chemical potentials.

If there are more than one axis, each exit created in a node will have an indication which phase should be held fix with zero amount along the line, which axis to use initially to vary the conditions and in which direction and a phase which is forbidden to become stable at the first axis increments. The last information is used to determine the sign of the first increment of the axis as shown in algorithm C1. There are loops in a phase diagram and one may eventually find the same node following a line, but not at the first increments.

All node points are stored in a list searched by algorithms C1 and C2. The lines share a **buffer** for sequential storage of all calculated equilibria along the lines.

3.2. Calculate diagrams with a single axis variable

Diagrams with equilibria calculated varying a single condition along an axis are useful to plot how various properties in the system varies with this condition. Such a step calculation can be used to represent many different types of property diagrams. The way to do such calculations is presented on the left branch of algorithm B in Fig. 4.

The user selects one of the conditions used for the initial equilibrium calculation as axis variable with a minimal and a maximal value as well as a maximum increment. Then a step command generates a node at the initial equilibrium with two exits, one in each direction of the axis. Algorithm C1 described in Fig. 5 will then calculate equilibria at each increment along the axis condition until it reaches the axis limits or the set of stable phases changes.

In the latter case algorithm C2 in Fig. 6 will handle the end of the current line and generate a node. In order to end the current line, the phase which appears or disappears is set fix with zero amount, the axis condition released and the equilibrium is calculated. Once this equilibrium has been calculated and checked, the line will be

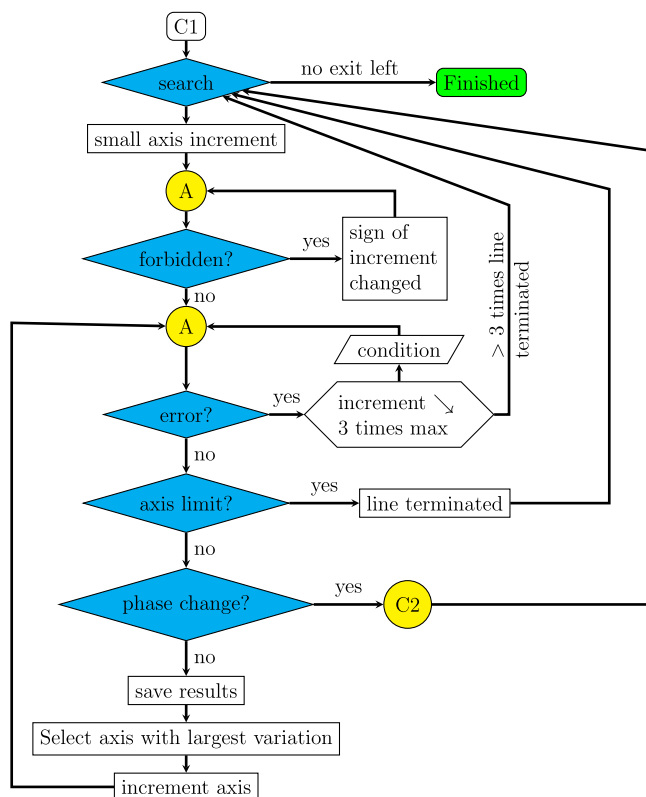


Fig. 5. Flowchart of algorithm C1 to follow equilibria along lines using algorithm A to calculate each equilibrium and call algorithm C2 when algorithm A finds a change of the set of stable phases.

terminated and a new node will be created with one exit to continue calculating along the axis in the same direction with the new set of stable phases. Algorithm C2 returns to algorithm C1 to calculate a new line. An example of plots after a step calculation is shown in Fig. 14.

3.3. Mapping with 2 axes

The same algorithms are used for calculating with one or two axis variables but the handling of the nodes is different because the nodes can contain more exits when there are two axes. The goal is to calculate all lines without duplication or forgetting any connected lines. The handling of the exits related to invariants in multicomponent systems is particularly tricky. As previously discussed, lines not connected to other lines is an issue that should be treated with extra starting points.

The mapping with two axes follows the branch on the right of algorithm B in Fig. 4. After an initial equilibrium calculation, two conditions are set as axes with a minimum and a maximum value and a maximum increment for each.

One of the axes, normally with a potential condition, is then incremented to calculate the equilibrium using algorithm A until the set of the stable phases changes. A first node point will be created with the appearing/disappearing phase as fix with zero amount and with two exits, one in positive and the other in negative direction of the other axis and algorithm C1 will be called.

Algorithm C1 will begin searching the list of nodes to find exits to generate lines. If there are none the mapping is finished. If it finds an exit it will generate the line, similarly to the step case discussed in Section 3.2, but in this case one phase will be fix with zero amount along the line, it will be a ZPF line. Algorithm C1 may increment either axis to follow the curvature of the line.

When starting a line from an exit, there is an initial axis to vary. It is first followed with a small increment in order to check that

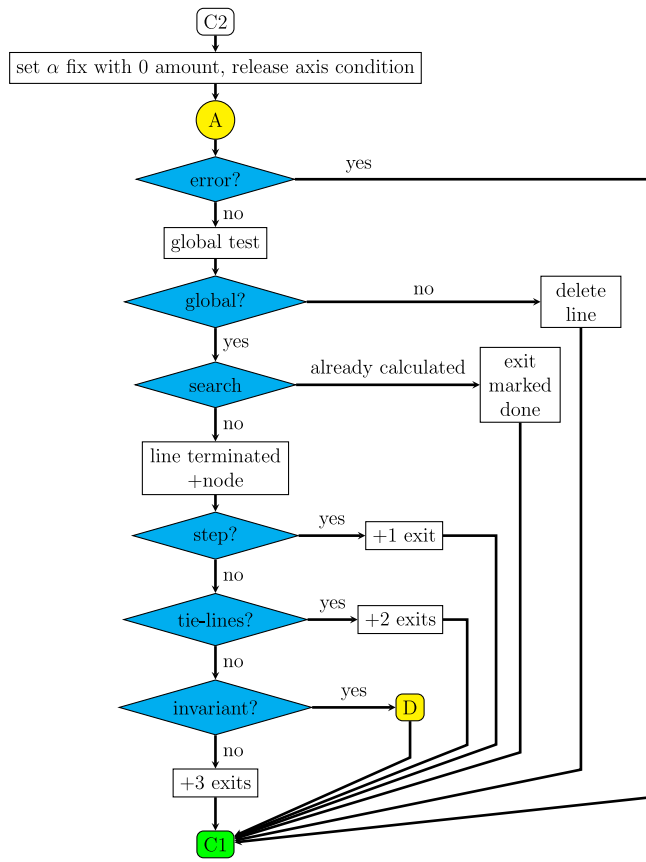


Fig. 6. Flowchart of algorithm C2 to handle a change of the set of stable phases at the end of a line and generate a node with exits for further lines.

the direction is correct. The axis condition will then be incremented between each call of the algorithm A to calculate single equilibria along the line. When an equilibrium has been properly calculated, it is stored in the buffer and algorithm C1 will check which axis varies most rapidly and possibly change the axis to use for incrementing the next iteration.

If algorithm A does not converge, the axis increment is decreased and the condition modified up to three times. If algorithm A has not converged after three times, the line is terminated and another exit is searched. The line is also terminated when the axis limit is reached.

The change of the set of stable phases is handled by algorithm C2. As in the case of stepping, in order to handle the change of line, algorithm C2 in Fig. 6 will first calculate the end point for the current line. This is done by fixing to zero amount the phase that wants to appear or disappear and release the current axis condition. The current line is terminated at this point and algorithm C2 will first check if the node has already been found. If so, an exit from the node, corresponding to the current line, is marked as done. Otherwise a new node point is created with one or more exits. In the rather simple cases where tie-lines are in the calculated plane, two exits will be created.

In iso-pleths, with extensive or normalized properties as conditions, in addition to T most node points correspond to two crossing lines as in Fig. 13(b). There are several such crossings in Fig. 13(a). Along each line a phase is fixed with zero amount and at the crossing two of the opposite regions has the same number of stable phases, in the other two opposite regions the number of stable phase differ by two. This is a characteristic feature of such iso-pleth sections. Such a node requires the creation of 3 exits when they are found.

Finally invariant equilibria require special attention. In Fig. 13(c) the two horizontal lines represent invariant equilibria with 6 stable phases in a 5 component isobaric system. Algorithm D, in Fig. 7,

explains how to define the exits from such nodes. An invariant node can be detected using the Gibbs phase rule:

$$f = n + 2 - p - c \quad (8)$$

where f is the degree of freedoms, n is the number of components and p is the number of stable phases. For each condition on a potential, which is not an axis variable, c is incremented by 1. At an invariant equilibrium $f = 0$. A binary isobaric phase diagram has $f = 3 - p$ in Eq. (8) and an invariant has thus 3 stable phases. An isobaric system with n elements has $p = n + 1$ number of stable phases at the invariant if T is an axis variable and no other conditions on potentials. Each stable phase at the invariant must have a ZPF line entering the invariant and another ZPF exiting the invariant, the maximum number of exits is thus equal to $2p$. Amongst these, algorithm D finds the ones in the section to be calculated.

At the calculated invariant equilibrium all phases have known compositions and other properties, it is only the amounts of phases that are different at the exits. We have 2 axes and $n - 1$ additional conditions on extensive or normalized variables. Two of the $n + 1$ phases stable at the invariant must have zero amount at each exit as they represent ZPF lines. We can use the remaining $n - 1$ phases and the $n - 1$ conditions to formulate a system of linear equations to obtain the phase amounts at each exit. This is done by summing up the contribution to the conditions from each phase multiplied with the unknown phase amount:

$$\begin{cases} \sum_{\alpha \neq \beta_1, \beta_2} \aleph^\alpha x_A^\alpha = \bar{x}_A \\ \vdots \\ \sum_{\alpha \neq \beta_1, \beta_2} \aleph^\alpha x_E^\alpha = \bar{x}_E \end{cases} \quad (9)$$

where $\bar{x}_A$ etc. are the conditions and the sum over α is for all stable phases excluding two, β_1 and β_2 , which are the two phases with ZPF lines at the exit. If any $\aleph^\alpha$ is negative solving this system of linear equations the exit is not in the calculated section.

The composition of the phases are known and also the conditions, the unknowns in Eq. (9) are the amount of phases, $\aleph^\alpha$. All the conditions on extensive or normalized properties in the plane of the calculation, not just mole fractions, must be included in Eq. (9).

At each exit, the compositions of the stable phases are the same but the amounts $\aleph^\alpha$ will be different and must be stored together with the phases β_1 and β_2 . Both of these have zero amount at the exit but along one line β_1 will be stable with zero amount and β_2 forbidden to become stable at the first increments and along the other line β_2 will be stable with zero amount and β_1 forbidden. This is tested by algorithm C1 in Fig. 5 at the first axis increment.

As explained in Fig. 7, this procedure has to be repeated on all the possible exits for the invariant i.e. looping over the pairs of phases stable at the invariant.

Algorithm D is illustrated in Fig. 8 in the simple case of an invariant in a ternary system. In Fig. 8(a), the composition of 4 phases, α , β , γ and δ , involved in an invariant equilibrium is represented at the invariant temperature together with the trace of the plane of the calculated iso-pleth, shown as a dashed red line.

In Fig. 8(b), the thick red line corresponds to the equilibria where γ and δ are excluded. It does not intersect with the calculated plane. No exit will be generated from this case. The thick green line corresponds to the equilibria where α and β are excluded. It intersects with the calculated plane in N_2 with positive values of amount for the phases γ and δ . A node will be created at this composition with two exits. These two exits can be better understood from Fig. 8(c) where an isothermal close to the invariant is represented. From the node, two lines will be calculated. The upper one follows the limit of the 3 phases field $\beta + \gamma + \delta$ where β is in zero amount while the lower one follows the limit of $\alpha + \gamma + \delta$ where α is in zero amount.

If the increment in T had the wrong sign when exiting the invariant in algorithm C1, the stable equilibrium from node N_2 would be $\alpha + \beta + \gamma$ with a positive amount of α and β . The equilibrium would be forbidden and the sign of the increment changed in order to calculate again at a T on the other side of the invariant.

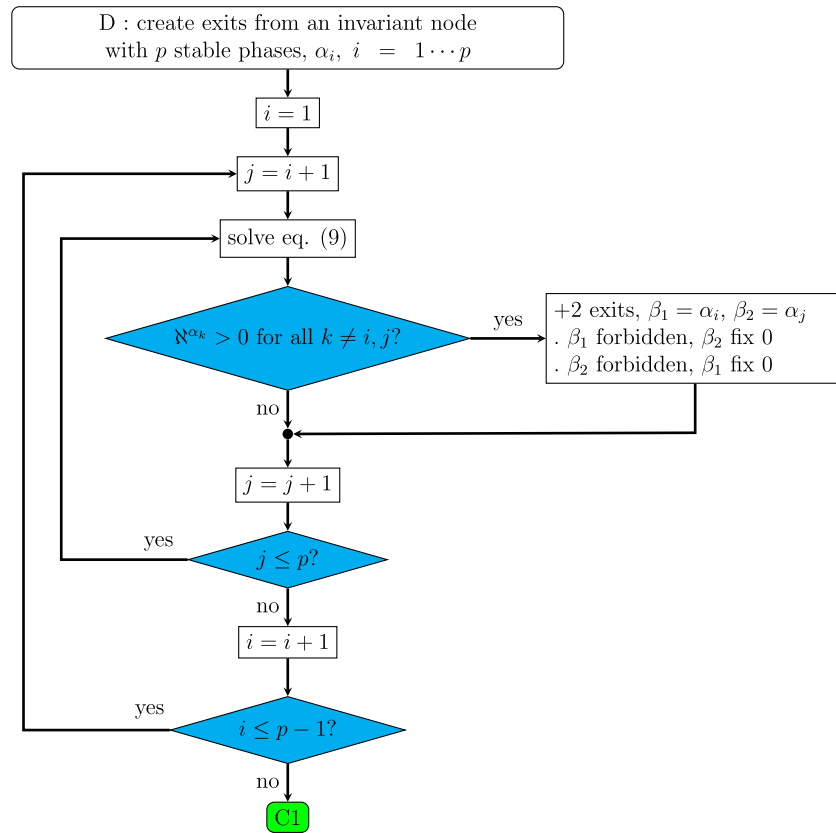


Fig. 7. Flowchart of algorithm D to find all exits in the plane of calculation from a multi-component invariant equilibrium i.e. to find all pairs α_i, α_j with zero amount together with positive amount of all other stable phases.

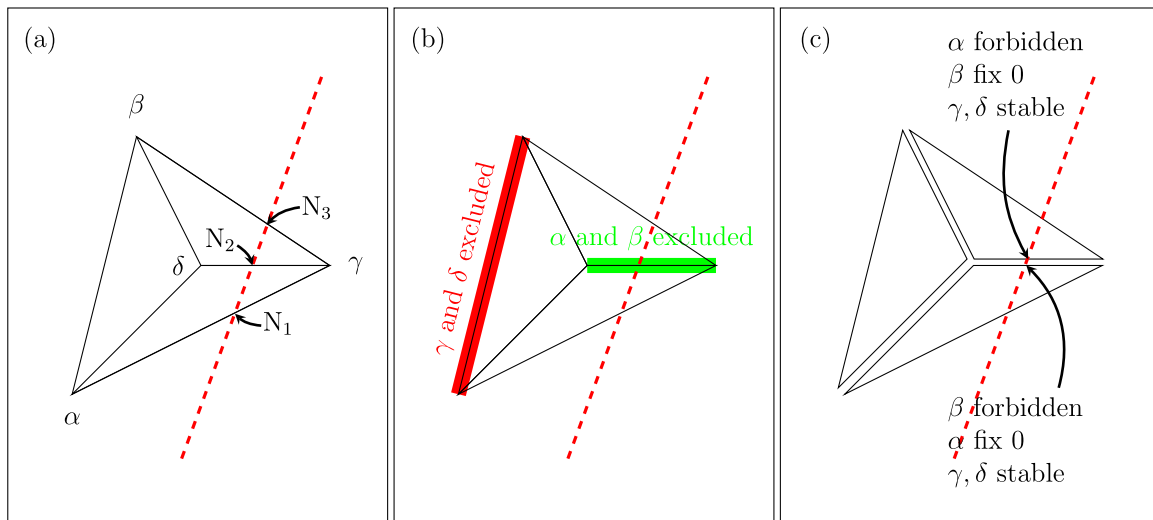


Fig. 8. Intersection of a ternary invariant with an iso-pleth section. In (a), the composition of the four phases taking part in the invariant equilibrium are shown at the invariant temperature together with a red dashed line showing the section being calculated. In that section, the invariant equilibrium will appear between the points N_1 and N_3 . In (b), in red when γ and δ are in zero amount there is no intersection with the calculated section but when α and β are in zero amount, the intersection of the green segment with the calculated section defines the node N_2 with positive amount of γ and δ . In (c), at a small increment in T , the two exits of N_2 are indicated with the status of the phases α and β .

3.4. Mapping explained line by line

An example how the mapping of a ternary iso-pleth progresses is shown in Fig. 9. This is also provided as screenshot in the supplementary material. The system is Fe–Cr–C with fixed 13% Cr by mass.

In Fig. 9, the iso-pleth is plotted after the calculation of each separate line. The initial conditions selected by the user at 1100 K and

1% C by mass gives an initial equilibrium where fcc and M_7C_3 are stable. Algorithm B decreases T until it finds a new phase stable, bcc in this case. This step is shown as a cyan line in the two first steps of Fig. 9.

Algorithm B will generate a node with two exits to follow this phase stable with zero amount in both directions of the C composition axis using algorithm C1. One of these lines ends at the axis limit in Fig. 9(a). In Fig. 9(b) the line is followed in the other direction until a new phase

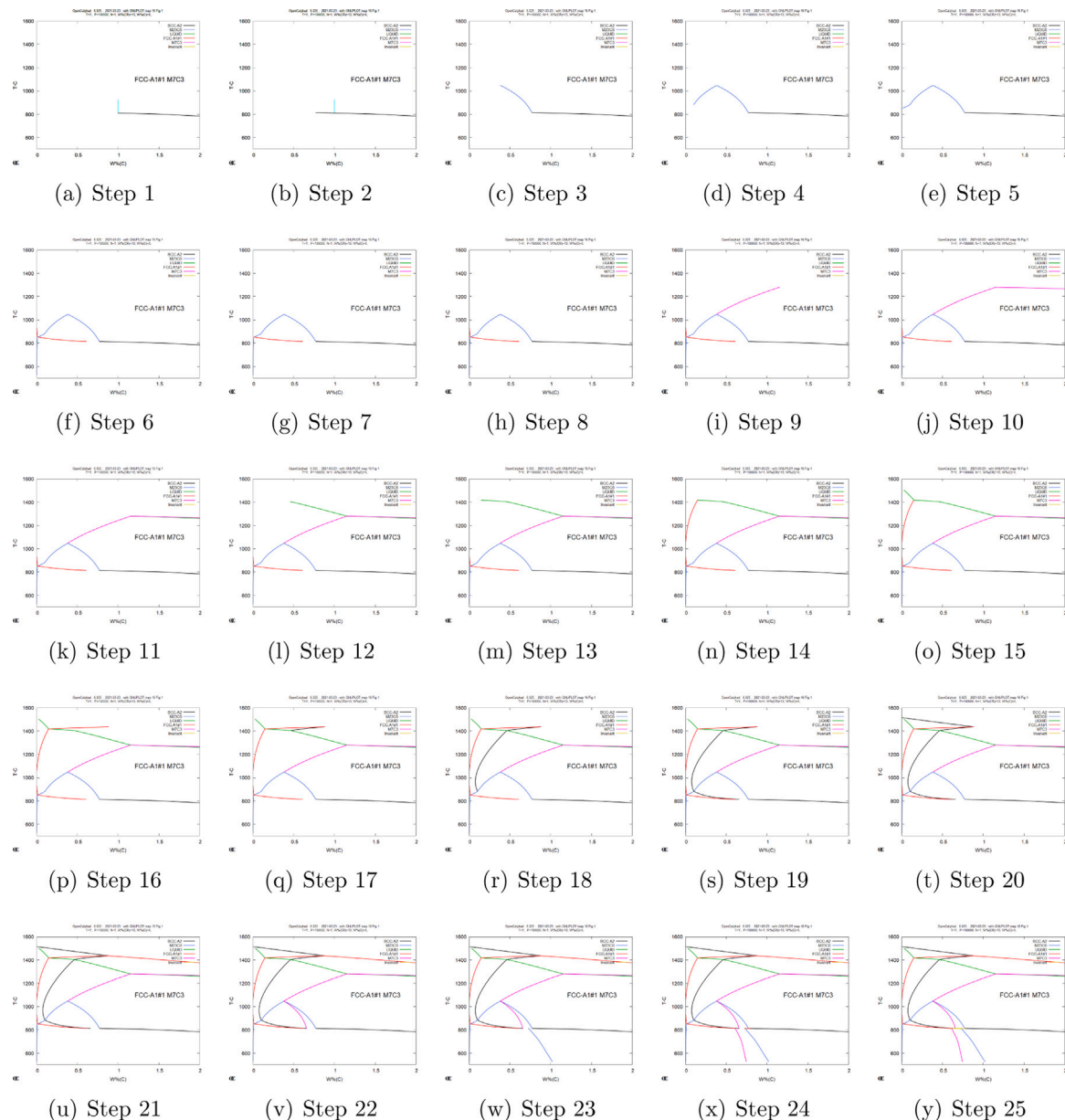


Fig. 9. Mapping the C-Cr-Fe system step by step. See the supplementary material for a detailed explanation of each step.

becomes stable, $M_{23}C_6$. This is an invariant equilibrium as we now have 4 stable phases. In the added material there is an explanation of all the further steps.

Each line ends at an axis limit or when there is a change of the set of stable phases. In algorithm C2 it is checked if the end of the line represents an already existing exit node and the corresponding exit is eliminated, otherwise a new node is created with new exits.

4. Plotting

After a calculation with varying conditions has been performed, OC has saved the results for all calculated equilibria. This means that the results can be plotted in many different ways, not just a traditional phase diagram. To understand the rules to plot a correct phase diagram please see Hillert [19]. Some examples are given here to show this feature. Some conditions or state variables can be quite meaningless as axis variables for step or map commands or as plot axes. Some other will generate strange diagrams but it is up to the user to decide what he or she is interested in.

4.1. Binary diagram

The Ag-Cu diagram shown in Fig. 2(a) was calculated using T and mole fraction of Cu, x_{Cu} , as axis variables but for the plotting many different axes can be used as shown in Fig. 10, where Fig. 10(a) is the same as Fig. 2(a). In (b) the overall composition in Cu is plotted on the horizontal axis. The lines corresponds to the overall composition and that does not include the composition of the phase fix with zero amount. But as OC saves complete descriptions of each calculated equilibrium along the lines, the composition of the phase with zero amount can be extracted and plotted as in Fig. 10(a) if the user selects the mole fraction of Cu in all stable phases as axis variable (which is the default when there are tie-lines in the plane).

All diagrams in Fig. 10, except for (b), are proper phase diagrams. An inexperienced user may obtain a diagram as (b) without intention. It is quite easy to detect that the proper axes have not been used in a binary case but could be undetected in more complex cases. In Fig. 10(c) the activity of Cu is plotted as horizontal axis. Thus the two

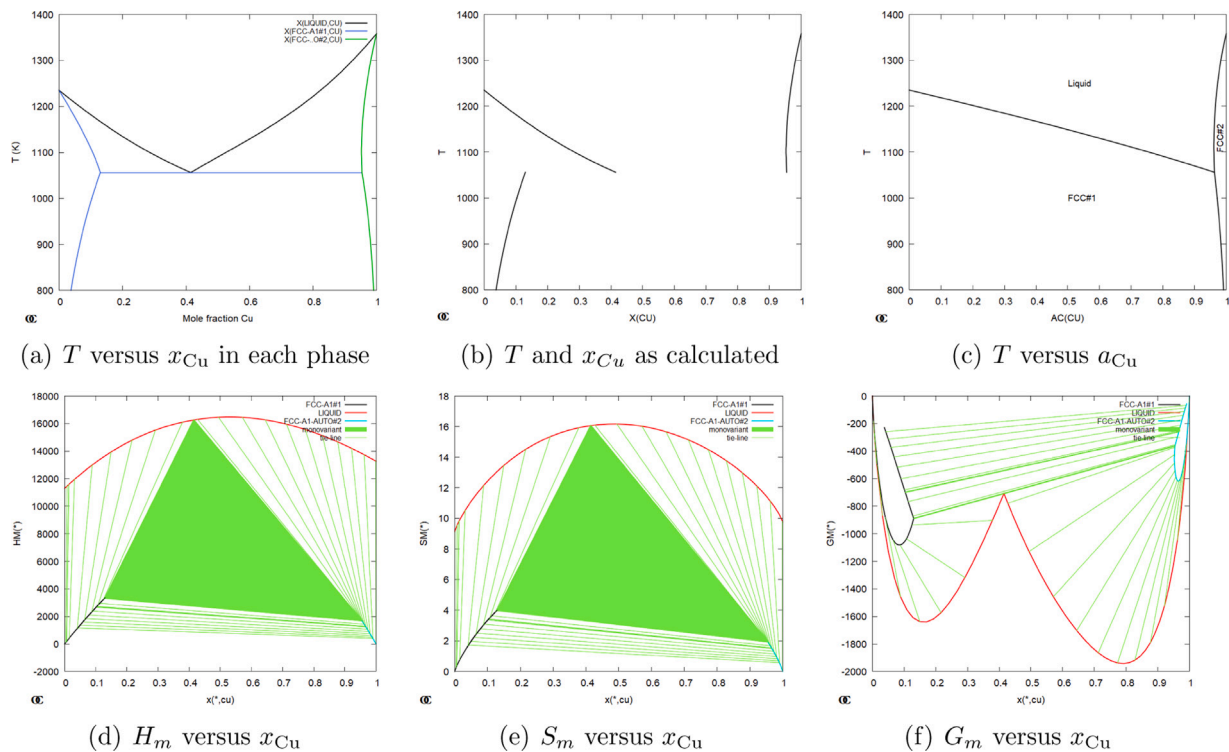


Fig. 10. The binary system Ag-Cu plotted with different axes from the same calculation, the reference state for the activity, enthalpy, entropy and Gibbs energy is the fcc phase at the current T , see the text for explanations. The system was assessed by Hayes et al. [15].

solubility lines becomes a single one and the invariant equilibrium is a point because the chemical potentials in the stable phases are the same.

In (d) the molar enthalpy of the different phases is plotted on the vertical axis and the diagrams becomes topologically similar to a ternary iso-thermal section with a large invariant area. Inside this area the amount of phases varies but all potentials are constant. It is plotted for the equilibrium that have been calculated during the mapping *i.e.* when three phases are in equilibrium. The molar enthalpy of the liquid phase is shown in red, the one for the Ag rich fcc phase in black and the one for the Cu rich fcc phase in cyan. For all the phases in this system, the mixing enthalpy is positive leading to a miscibility gap at low temperature in the fcc phase. In the description used to make the calculation, the enthalpy of mixing of the solid phases does not depend on temperature; their range of stability is thus a line. The tie lines, shown in green, connect the composition of the phases in equilibrium. In Fig. 10(a), they would appear as horizontal lines connecting phases in equilibrium at the same T . The green triangle corresponds to the invariant equilibrium. Its lower edge corresponds to the solid state at the end of the eutectic reaction. Any point in this triangle would represent a state of the ongoing eutectic invariant reaction. The melting enthalpy at a given composition can be estimated from the difference between the red line and the solid state at the solidus *i.e.* the black line, the lower edge of the triangle or the cyan line at the composition under consideration.

In (e), a similar diagram with the molar entropy of the phases on the vertical axis is shown. In (f), the molar Gibbs energy of each phase is plotted on the vertical axis. As the G functions vary with temperature, the fcc single phase areas are not any more a single line but a portion of the plane between the solidus for lower G values connected with the red line and the solvus connected to the other end solution. Extending the tie-lines in the two-phase area to the vertical axis allows to read the chemical potential for each element. The invariant is reduced to a line; the chemical potential of both elements is the same in all three phases at the invariant equilibrium. Explaining a binary phase diagram with these figures may provide some additional understanding of thermodynamic relations.

4.2. Ternary iso-pleth

In Fig. 11(a), the phase diagram generated after the mapping explained in detail in Fig. 9 is shown. The labels linked to the color of the lines allow to know which phases are stable in the different regions. It is however impossible from this diagram to read the composition of the phases and their amount. The two iso-thermal diagrams plotted in Fig. 11(b) and 11(c) can provide this information at the corresponding temperatures. The temperatures corresponding to these iso-thermal sections are indicated as horizontal dotted lines in Fig. 11(a). The compositions in the iso-pleth section are shown in the iso-thermal sections by a dotted red segment. In Fig. 11(b) and 11(c), the tie-lines are drawn as light green lines. The large green areas correspond to tie-triangles *i.e.* areas where three phases are stable. The tie-lines and tie-triangles intersecting with the red dotted line allow to better understand the equilibria in the iso-pleth section.

At 900 °C, the fcc phase is present in major amount all along the section. Actually, in a small range between the black and the blue lines close to 0.1% C, it is even present as a single phase. The other phases stable in the iso-pleth are carbides rich in chromium. They are present in small phase amounts. In the low C content area, the bcc phase becomes stable. Its composition is close to the section and thus its phase fraction is more important, close to the fcc one.

At 1427 °C, the composition of the phases in equilibrium in the iso-pleth section are all close to the section. The liquid phase is stable as single phase for the higher C content. When the C content decreases the system is first two-phase fcc+liquid, then three-phase bcc+fcc+liquid, then bcc+liquid and finally single-phase bcc.

In Fig. 12(a), we have the same phase diagram as in Fig. 11(a) *i.e.* using the overall content of C in weight percent, $\%w_C$, as x axis. In Fig. 12(b), the C weight percent of the stable phases, $\%w_C^*$, is plotted as x axis in the same range of composition as the overall composition. Only the composition of the metallic phases is shown as the carbide phases are outside the diagram with much higher content in C. This means that not all lines have 13% Cr which was the condition for the

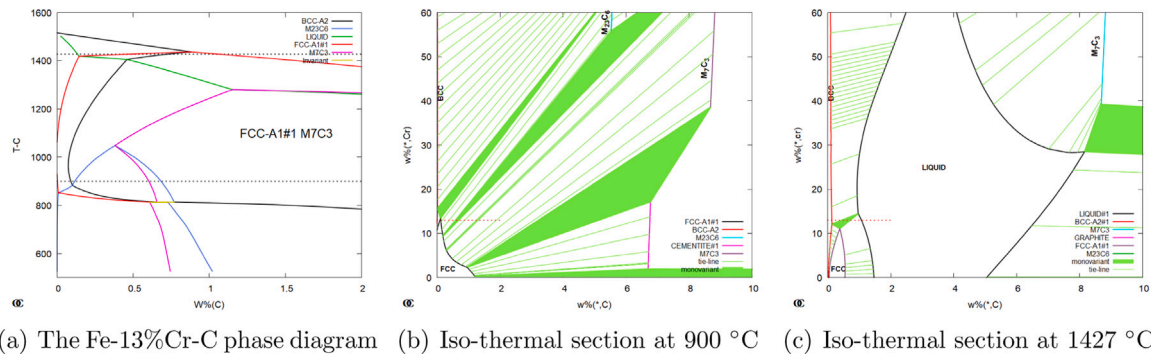


Fig. 11. The C–Cr–Fe system calculated as an iso-pleth and as two iso-thermal sections. The two dashed lines in (a) indicate the T for the iso-thermal sections. In these there is a red dashed line at 13%Cr from 0 to 2% C representing the iso-pleth. Equilibria with higher carbon phases have been suppressed.

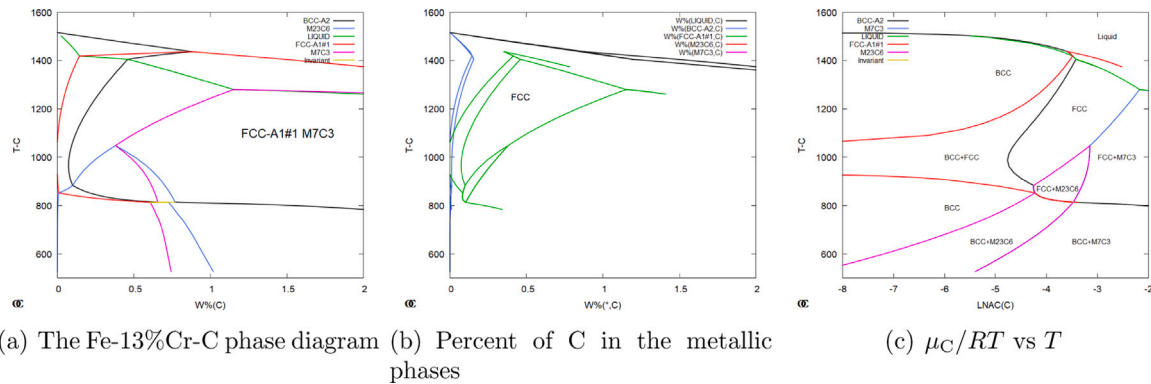


Fig. 12. In (a) the same C–Cr–Fe system as in Fig. 11(a) and in (b) the C composition in the metallic phases as calculated along the lines. The Cr content is no longer 13% except around the regions with a single fcc, bcc or Liquid phase. It is not a phase diagram but plotted from same calculation and it is instructive to figure out which phases have been stable along each line. In (c) the horizontal axis in μ_C/RT (with graphite as reference state).

mapping. The lines around the single phase regions of liquid, fcc and bcc in Fig. 12(b), are the same as in Fig. 12(a), because the phases have 13% Cr in equilibrium with another phase with zero amount. The other lines are when the phase itself has zero or varying amount in equilibrium with other phases. It is interesting to note that we can see the extent in T of the γ loop for the binary Cr–Fe system as the endpoints of the lines at zero C. This diagram is not a phase diagram but it could easily be obtained from a software by a user with or without intention. An advanced or careful user may find it interesting but most users will have difficulties in understanding it and disregard such a diagram.

In Fig. 12(c) the results of the same calculation are plotted using the chemical potential of C, relative to graphite, as horizontal axis. Some of the lines do not extend to the plot axis because the chemical potential for the calculated end point at 2% C per mass varies with temperature. But it is a phase diagram because there is a unique set of stable phases in each region although some of the lines, notably those separating regions with two stable phases, are not separated by a three-phase region as in Fig. 12(a). The reason is that the chemical potential of C is determined by the three-phase equilibrium with a common phase, either bcc, fcc or a carbide. The invariant 4-phase equilibrium is a point shared by the 4 two-phase regions.

4.3. Iso-pleth section in a complex system

When increasing the number of components, phase diagrams may become more complex and difficult to understand. In Fig. 13 an iso-pleth section for a five component High Speed Steel (HSS) is presented. It has 5% Cr, 8% Mo, 1% V by mass, rest Fe and C. The lines separate regions with different sets of stable phases and tie-lines cannot be drawn because at least one of the phase compositions is outside the

plane of the diagram. Fig. 13(b) is a zoom in the area around 1600 K and low C where there is a node involving three phases: the liquid, bcc and fcc. Four lines are intersecting at this node. In Fig. 13(c) some of the invariants at low T are shown, they have constant T and each has 6 stable phases and in these cases 8 exits. There is an invariant at higher T with just 6 exits in the plane of the diagram.

To obtain more detailed information on the equilibria in this section, one can use step calculations, as shown in Fig. 14, or calculate single equilibria.

4.4. Step calculation

It is useful to complement the phase diagram in Fig. 13 with additional single equilibria or variations of a single axis to understand a complex material under consideration.

For this example, we have calculated a “step” diagram at fixed content of C, 0.8% per mass, and varied T . In Fig. 14, three diagrams from this calculation have been plotted. In (a), the phase fractions show that a significant amount of bcc is formed from the liquid but quickly replaced by fcc in accordance with Fig. 13(a). Just before the liquid disappears, the M_6C carbide forms and later in the solid state the cubic MC carbide is formed. This phase has the same structure and model as the fcc phase but contains mainly V and C as shown in (b) plotting the V content in all stable phases. Several more carbides are formed at lower T and finally all fcc transforms to bcc which is the stable form of Fe at low T . Another property, the heat capacity of the alloy, is plotted in (c) as function of T , including the latent heat from the phase transformations.

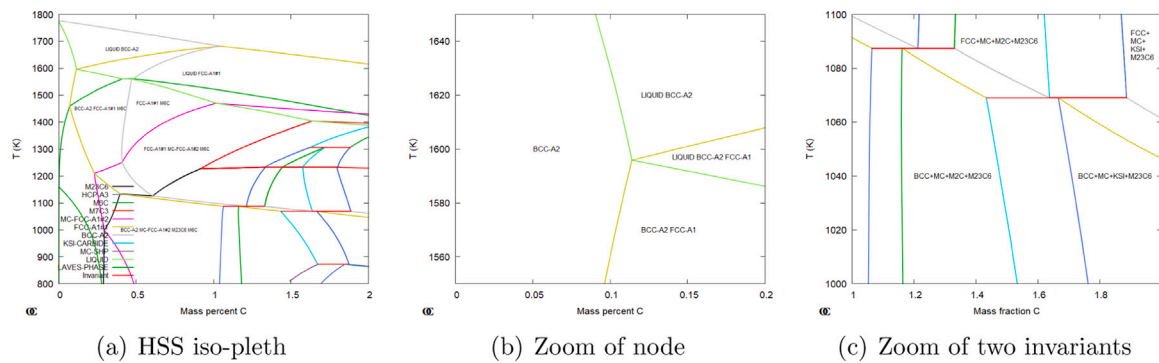


Fig. 13. An iso-pleth for a 5 component HSS. In (b) a zoom of a node point 1600 K and low C. At the lower invariant in (c) there are 6 stable phases: fcc+bcc+MC+M2C+KSI+M23C6. Calculated from the SGTE SSOL database [16].

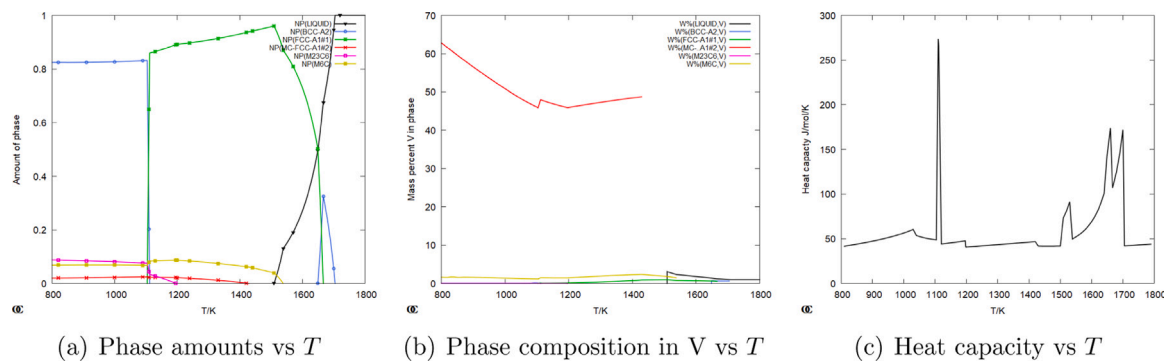


Fig. 14. Examples of property diagrams plotted after a step calculation in the same HSS as in Fig. 13 for constant content of C, 0.8% per mass. In (a) the phase amounts, in (b) the percent V in different phases and in (c) the heat capacity including latent heat at the phase changes. Using the SGTE SSOL database [16].

5. Summary

The main intention of this paper was to explain how one can calculate ZPF lines in multi-component systems. It is possible thanks to the flexibility of the use of Lagrange multiplier method in the calculation of single equilibria where most of the computation time is spent.

The diagrams show the power of the tools in Computational Thermodynamics. Traditional phase diagrams can be obtained but much more important is the possibility of detailed study how a material of interest is affected by varying the actual composition and heat treatment. Some of the diagrams may require extra effort to be understood but most software include more tools for calculations than explained here. The algorithms presented here can be extended to calculate ZPF lines with 3 or more axes. In such a case there will be 2 or more phases fix with zero amount along each line.

Thermodynamic software and databases are useful for planning experimental work and also needed for simulations of phase transformations together with kinetic data and models, see for example [20–26] because it is important that the thermodynamic databases used for simulations also reproduce the equilibrium state.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.calphad.2021.102330>.

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